

INTERNATIONAL MATHEMATICAL OLYMPIAD — 1992

FIRST DAY

1. Find integers a, b, c with $1 < a < b < c$ such that $(a-1)(b-1)(c-1)$ divides $abc-1$.
2. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x^2 + f(y)) = y + (f(x))^2$.
3. Nine points in space, no four coplanar, edges colored red/blue/uncolored. Find minimum colored edges guaranteeing a monochromatic triangle.

SECOND DAY

4. Circle C , tangent line L , point M on L . Find locus of P such that C is incircle of triangle PQR where M is midpoint of QR .
5. For finite set S in space with projections S_x, S_y, S_z , prove $|S|^2 \leq |S_x||S_y||S_z|$.
6. Define $S(n)$ = largest k such that n^2 is sum of k positive squares.
 - (a) Prove $S(n) \leq n^2 - 14$ for $n \geq 4$.
 - (b) Find n with equality.
 - (c) Show infinitely many such n exist.

INTERNATIONAL MATHEMATICAL OLYMPIAD — 1993

1. Let $f(x) = x^n + 5x^{n-1} + 3$. Prove it is not product of two nonconstant integer polynomials.
2. Triangle geometry condition with point D . (a) Find $(AB \cdot CD)/(AC \cdot BD)$. (b) Prove certain tangents are perpendicular.
3. Chessboard jumping game with n^2 pieces. Determine n for which one piece can remain.

SECOND DAY

4. For points define $m(PQR) = \text{minimum altitude}$. Prove $m(ABC) \leq m(ABX) + m(AXC) + m(XBC)$.
5. Does there exist $f : \mathbb{N} \rightarrow \mathbb{N}$ with $f(1) = 2$, $f(f(n)) = f(n) + n$, strictly increasing?
6. Circular lamps switching process. Prove cycle length formulas for even/odd n .

INTERNATIONAL MATHEMATICAL OLYMPIAD — 1994

1. Distinct a_i in $\{1, \dots, n\}$ closed under sums $\leq n$. Prove average $\geq (n+1)/2$.
2. Isosceles triangle configuration. Prove $OQ \perp EF$ iff $QE = QF$.
3. Let $f(k)$ count numbers in $\{k+1, \dots, 2k\}$ with exactly three 1s in binary.
 - (a) Show every m occurs.
 - (b) Find m with unique k .
4. Find all (m, n) such that $(n^3 + 1)/(mn - 1)$ is integer.
5. Find $f : S \rightarrow S$, $S > -1$, satisfying functional equation and monotonic ratio condition.
6. Show existence of integer set A with prime-product property.